DATABASE MANAGEMENT SYSTEMS

Database Security

2025

Copyright © 1995, 2024, Oracle and/or its affiliates

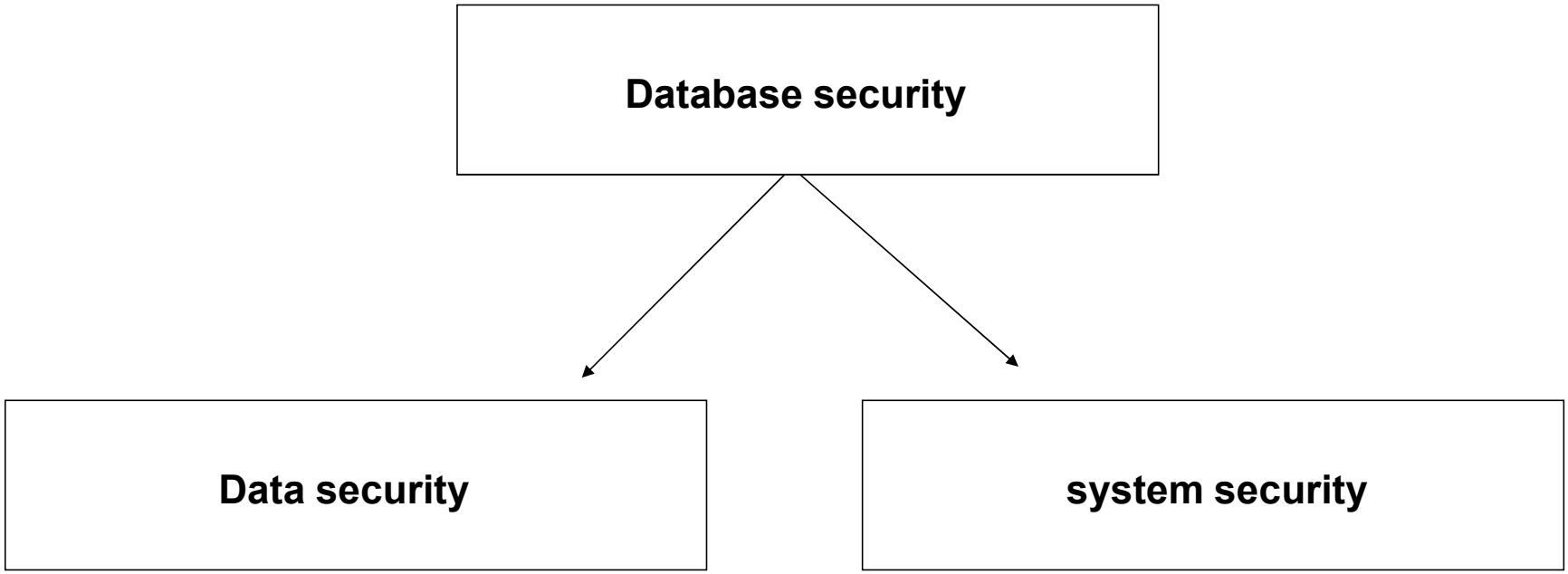
Database Security

Multi-user database systems like **Oracle** include security to control how the database is accessed and used for example security Mechanisms:

- Prevent unauthorized database access
- Prevent unauthorized access to schema objects
- Control disk usage
- Audit user actions

Copyright © 1995, 2024, Oracle and/or its affiliates

Database Security



Database Security

System Security:

Covers access and use of the database at the system level, such as:

- The username and password
- The disk space allocated to users,
- And the system operations that users can perform

Copyright © 1995, 2024, Oracle and/or its affiliates

Database Security

Data security :

Covers access and use of the database objects and the actions that those users can have on the objects such as:

- Selecting data from a table or retrieving a value from a sequence

Copyright © 1995, 2024, Oracle and/or its affiliates

Database Security

User Security (AAA security model)

- Authentication: verifying the identity of someone (a user, device, or an entity) who wants to access data, resources, or applications.
  - It give us initial access & many authentication methods.

- Authorization: Access limits for authenticated users.
  - Least privileges principle: user must be able to access only the information and resources that are necessary for its legitimate purpose and no more

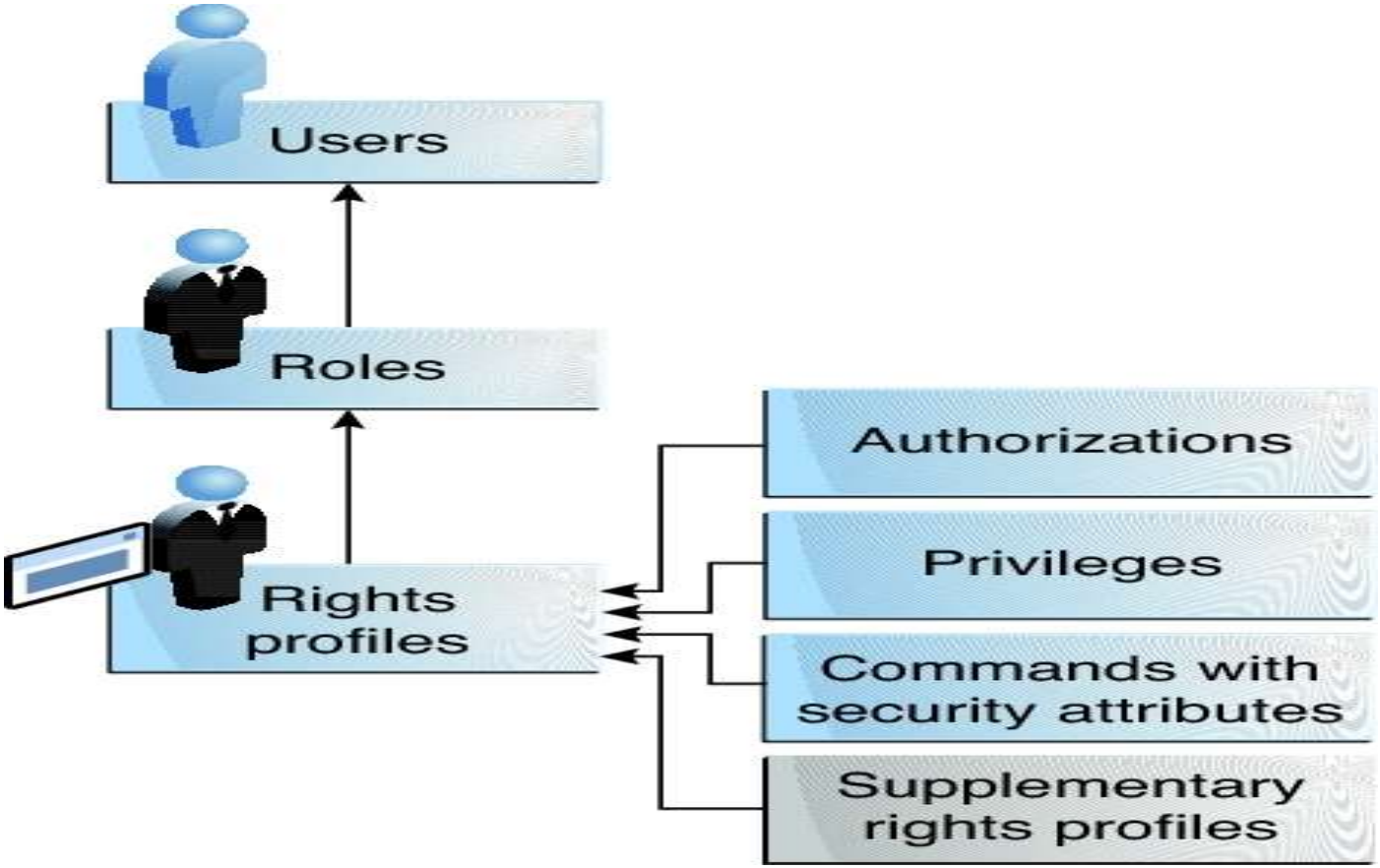
- Accounting: auditing

Copyright © 1995, 2024, Oracle and/or its affiliates

Database Security

User Security

- DB Account
- Privileges
- Roles
- profiles



Copyright © 1995, 2024, Oracle and/or its affiliates

Database Security

User Accounts

- Default user accounts:
  - The following administrative user accounts are automatically created when you install Oracle Database
    - **SYS**: granted by DBA role, owns Data Dictionary
    - **SYSTEM**: granted the DBA role. This is the user account that you log in with to perform all administrative functions, except:

      1. Startup and shutdown database
      2. Backup database

Copyright © 1995, 2024, Oracle and/or its affiliates

Database Security

Components of an oracle user account

- **Unique username** : it should be less than 30 character, contains alphanumeric, $ and _
  - You can not use keyword. E.g. insert, create … etc.
- **Authentication method**: password, OS or biometrics
- **Default tablespace** to store data
  - It will be set to SYSTEM if not specified
- **Temporary tablespace** for query processing work space
- **Profile**: It allows you to regulate the amount of resources used by each database user & enforce complexity of password
  - To apply specific setting to a group of users. First create a profile then assign users to it.
- **account status** : it can be: open, locked(attempts) or expired

Copyright © 1995, 2024, Oracle and/or its affiliates

Creating Users

- The DBA creates the user by executing the CREATE USER statement.
- The user does not have any privileges at this point.
- The DBA can then grant privileges to that user.
- These privileges determine what the user can do at the database level.

Copyright © 1995, 2024, Oracle and/or its affiliates

Creating Users

The syntax for creating a user is:

```
CREATE USER user
IDENTIFIED BY   password
DEFAULT   TABLESPACE   tablespace_name
TEMPORARY    TABLESPACE  tablespace_name
QUOTA [UNLIMITED ] [integer M] on tablespace_name
PROFILE profile_name;
```

Example:

```
CREATE USER Scott
IDENTIFIED BY   tiger
DEFAULT TABLESPACE system
TEMPORARY    TABLESPACE temp
QUOTA UNLIMITED on system
PROFILE managerProfile;
```


Copyright © 1995, 2024, Oracle and/or its affiliates

Creating Users

Changing Your Password

- ✓ The DBA creates your user account and initializes **your password.**
- ✓ You can change your password by using the **ALTER USER statement.**

Syntax :

```
ALTER USER user IDENTIFIED BY newpassword;
```

Example : :

```
ALTER USER Scott  IDENTIFIED BY lion;
```


Copyright © 1995, 2024, Oracle and/or its affiliates

Privileges

Privileges are the right to execute particular SQL statements. The database administrator (DBA) is a high-level user with the ability to grant users access to the database and its objects

Example:

- The ability to connect to the database
- The ability to create a user
- The ability to create a table

Copyright © 1995, 2024, Oracle and/or its affiliates

Privileges Types

- **System privileges**: Gaining access to the database
- **Object privileges**: Manipulating the content of the database objects

Copyright © 1995, 2024, Oracle and/or its affiliates

Privileges

USER PRIVILEGE

SYSTEM PRIVILEGE

Permission to perform any action on any schema objects such as ALTER Or CREATE TABLESPACE, CREATE USER, ALTER USER, DROP USER, ALTER SYSTEM

```
SQL> Grant create table, create view to salesrep;

Grant succeeded.
```

OBJECT PRIVILEGE

Grant permission to a user on a database object. For example update or delete table, Execute procedure, function, views of another user.

```
SQL> Grant select on hr.company to salesrep;

Grant succeeded.
```

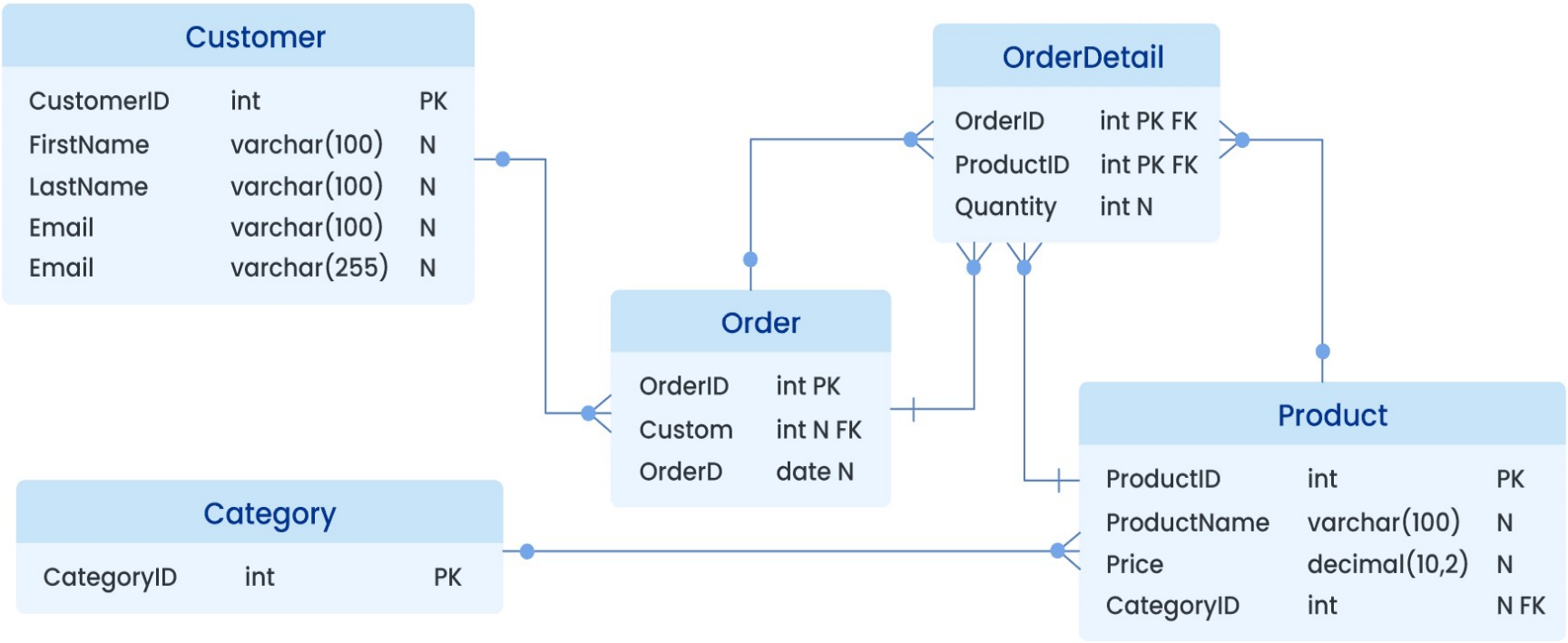

Copyright © 1995, 2024, Oracle and/or its affiliates

schema

- A schema is a collection of objects, such as tables, views, and sequences.
- The schema is owned by a database user who create it and has the same name as that user.

Copyright © 1995, 2024, Oracle and/or its affiliates

schema


Customer
CustomerID int PK
FirstName varchar(100) N
LastName varchar(100) N
Email varchar(100) N
Email varchar(255) N
OrderDetail
OrderID int PK FK
ProductID int PK FK
Quantity int N
Order
OrderID int PK
Custom int N FK
OrderD date N
Product
ProductID int PK
ProductName varchar(100) N
Price decimal(10,2) N
CategoryID int N FK
Category
CategoryID int PK

System Privileges

System privileges can be given to a user by another user who has administrator privileges or by a user who has the right to grant a system privilege.

More than 200 **privileges** are available such as:

- Creating new users
- Removing users
- Removing tables
- Backing up tables

Copyright © 1995, 2024, Oracle and/or its affiliates

System Privileges

Special Administrative privileges: required for an administrator to perform basic database operations are granted through two special system privileges

- **SYSDBA** privilege: can do anything
- **SYSOPER** privilege: sub-admin access, can perform:
  - Backup, recover, startup, shutdown
  - No access to data itself

Copyright © 1995, 2024, Oracle and/or its affiliates

System Privilege	Operations Authorized
CREATE USER	Grantee can create other Oracle users (a privilege required for a DBA role).
DROP USER	Grantee can drop another user.
DROP ANY TABLE	Grantee can drop a table in any schema.
BACKUP ANY TABLE	Grantee can back up any table in any schema with the export utility
CREATE ANY TABLE	Grantee can create tables in any schema.
SELECT ANY TABLE	Grantee can query tables, views, or snapshots in any schema

Copyright © 1995, 2024, Oracle and/or its affiliates

User Privileges

User System Privileges

Once a user is created, the DBA can grant specific system privileges to a user.

```
GRANT privilege
TO user [WITH ADMIN OPTION] ;
```

WITH ADMIN OPTION: it means give grantee right to grant the same privileges to other users

Copyright © 1995, 2024, Oracle and/or its affiliates

User Privileges

An application developer, for example, may have **the following system privileges:**

- CREATE SESSION
- CREATE TABLE
- CREATE SEQUENCE
- CREATE VIEW
- CREATE PROCEDURE

Copyright © 1995, 2024, Oracle and/or its affiliates

User Privileges

Granting System Privileges

- The DBA can grant a user specific system privileges.
- Example:

```
GRANT create session, create table, create sequence, create view

TO Scott;
```


Copyright © 1995, 2024, Oracle and/or its affiliates

Object Privileges

USER PRIVILEGE

SYSTEM PRIVILEGE

Permission to perform any action on any schema objects such as ALTER Or CREATE TABLESPACE, CREATE USER, ALTER USER, DROP USER, ALTER SYSTEM

```
SQL> Grant create table, create view to salesrep;

Grant succeeded.
```

OBJECT PRIVILEGE

Grant permission to a user on a database object. For example update or delete table, Execute procedure, function, views of another user.

```
SQL> Grant select on hr.company to salesrep;

Grant succeeded.
```


Copyright © 1995, 2024, Oracle and/or its affiliates

Object Privileges

Object privilege

- Is a privilege or right to perform a particular action on a specific (object) table, view, sequence, or procedure

- Each object has a particular set of grantable privileges. The table in the next slide lists the privileges for various objects

Copyright © 1995, 2024, Oracle and/or its affiliates

Object Privileges

Object Privilege	Table	View	Sequence	Procedure
ALTER	√		√	
DELETE	√	√		
EXECUTE				√
INDEX	√			
INSERT	√	√		
REFERENCES	√	√		
SELECT	√	√	√	
UPDATE	√	√		

Copyright © 1995, 2024, Oracle and/or its affiliates

Object Privileges

- Object privileges vary from object to object.
- An owner has all the privileges on the object.
- An owner can give specific privileges on that **owner's object.**

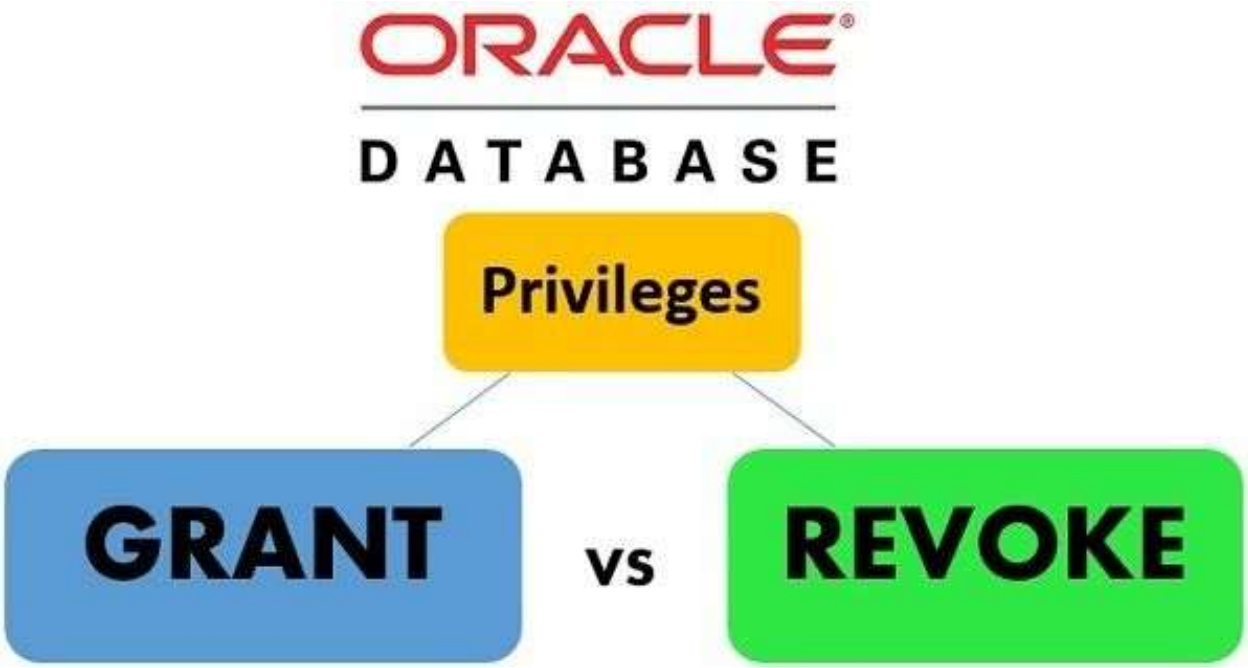
Syntax:

```
GRANT object_privilege [(columns)]
ON object
TO user
[WITH GRANT OPTION];
```

If the grant includes **WITH GRANT OPTION**, then the grantee can further grant the object privilege to other users; otherwise, the grantee can use the privilege but cannot grant it to other users.

Copyright © 1995, 2024, Oracle and/or its affiliates

Granting Object Privileges



Granting Object Privileges

- **Grant** query privileges on the **EMPLOYEES** table.

```
GRANT  select
ON     employees
TO     Norah, Sarah;
```

- **Grant** privileges to update specific columns to **users and roles.**

```
GRANT  update (department_name, location_id)
ON     departments
TO Scott, manager;
```


Copyright © 1995, 2024, Oracle and/or its affiliates

Revoke Object Privileges

- Remove privileges granted to other users by using the REVOKE statement.

- When you use the REVOKE statement you prevent the user from doing specific actions depending on the privileges you revoke from the user.

Copyright © 1995, 2024, Oracle and/or its affiliates

Revoke Object Privileges

Syntax:

```
REVOKE privilege ,ALL
ON     object
FROM   user, role, PUBLIC;
```

Example:

```
REVOKE  select, insert
ON      departments
FROM    Scott;
```


Copyright © 1995, 2024, Oracle and/or its affiliates

Revoke Object Privileges

- Privileges granted to others through the WITH **GRANT OPTION clause are also revoked**

- For example, if user A grants SELECT privilege on a table to user B including the **WITH GRANT OPTION** clause, user B can grant to user C the SELECT privilege with the WITH GRANT OPTION clause as well, and user C can then grant to user D the SELECT privilege. If user A revokes privilege from user B, then the privileges granted to users C and D are also revoked.

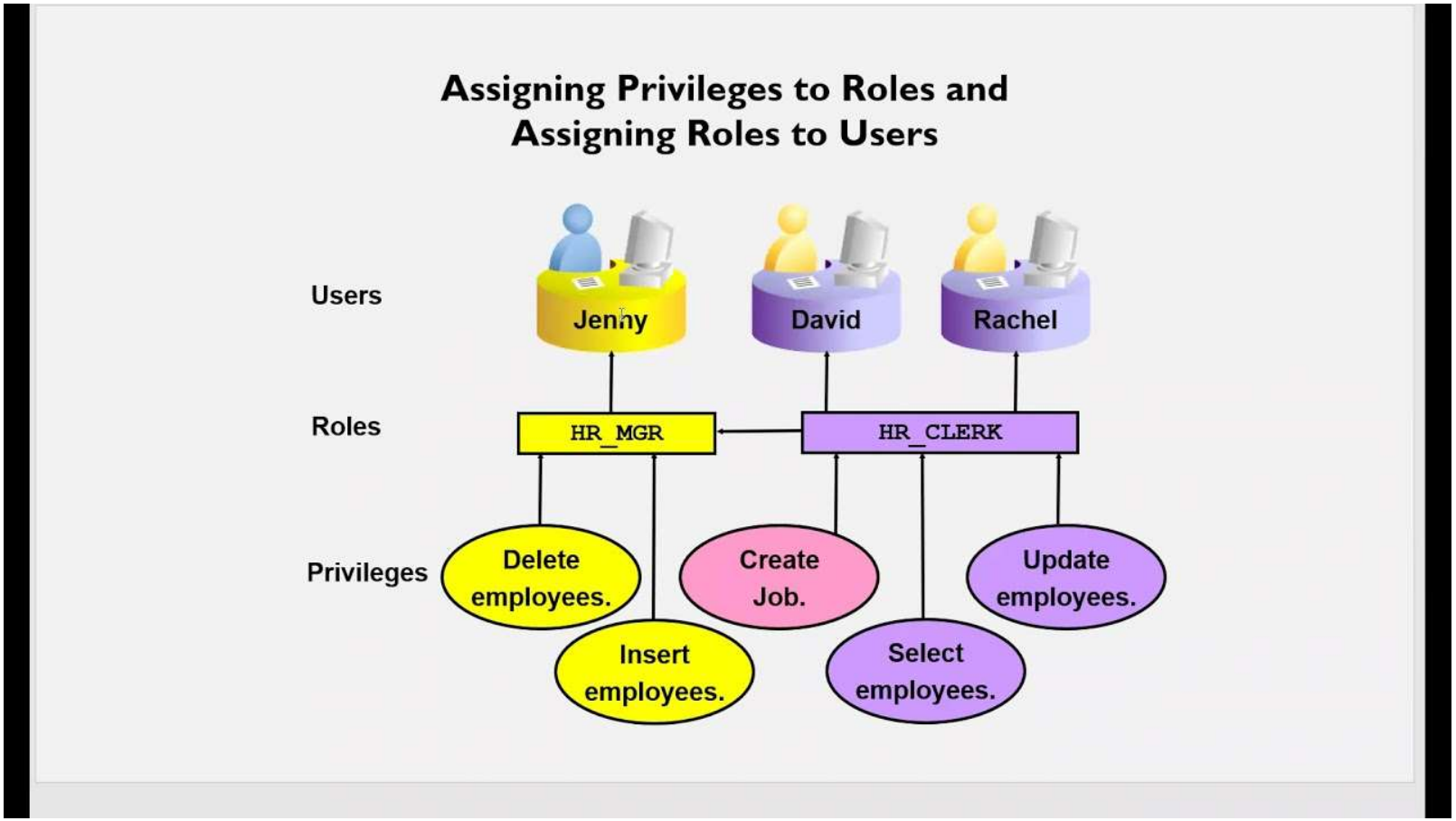
Copyright © 1995, 2024, Oracle and/or its affiliates

Roles

- A role is a named group of related privileges that can be granted to the user.
- This method makes it easier to revoke and maintain privileges.
- user can have access to several roles, and several users can be assigned the same role

Copyright © 1995, 2024, Oracle and/or its affiliates

Roles



Copyright © 1995, 2024, Oracle and/or its affiliates

Roles

Pre-defined roles:

- ✓ **DBA**: it has all system privileges (which SYS/SYSTEM have)
- ✓ **RESOURCE**: Enables a user to create certain types of objects in his own schema
- ✓ **CONNECT**: Enables a user to connect to the database. Grant this role to any user or application that needs database access.

Copyright © 1995, 2024, Oracle and/or its affiliates

Roles

Creating and Assigning a Role

- First, the DBA must create the role.
- Then the DBA can assign privileges to the role and users to the role.

Syntax :

```
CREATE   ROLE   role;
```


Copyright © 1995, 2024, Oracle and/or its affiliates

Roles

Creating and Granting Privileges to a Role

Create a role

CREATE ROLE manager;

Grant system privileges to a role

GRANT create table, create view
TO manager;

Grant a role to users

GRANT manager TO Maha, Nora;

Copyright © 1995, 2024, Oracle and/or its affiliates

User Security Guidelines

User Security Guidelines

- ✓ To grant privileges on an object, the object must be in your own schema, or you must have been granted the object privileges **WITH GRANT OPTION** .
- ✓ An object owner can grant any object privilege on the object to any other user or role of the database.
- ✓ The owner of an object automatically acquires all object privileges on that object.
- ✓ Do not give your users more abilities than they need to get the job done.

Copyright © 1995, 2024, Oracle and/or its affiliates

User Security Guidelines…..cont..

- ✓ Expire and lock unnecessary users.

```
ALTER USER tiger  ACCOUNT LOCK ;
```


Copyright © 1995, 2024, Oracle and/or its affiliates

END

Copyright © 1995, 2024, Oracle and/or its affiliates